

Choosing and Combining Biological Foundation Models for Sequence Prediction

Literature Review and Methodological Framework

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1. Introduction, Problem Statement, and Advance Organizer

A foundational challenge of modern computational biology is predicting downstream cellular phenotypes—such as messenger RNA (mRNA) degradation rates and steady-state tissue expression—directly from primary nucleotide and amino acid sequences. Historically, researchers approached this problem through engineered biophysical heuristics, quantifying codon usage biases, secondary structure thermodynamics, and local sequence motifs (Presnyak et al. 1111–14; Agarwal and Kelley 245). Over the past three years, the field has undergone a paradigm shift driven by biological foundation models (BFMs). Leveraging self-supervised learning algorithms adapted from natural language processing, these large-scale transformer and state-space architectures have been trained on millions of unannotated genomic, transcriptomic, and proteomic sequences, learning dense numerical representations of biological grammar (Dalla-Torre et al. 287–89; Lin et al. 1123–26).

However, while unimodal foundation models excel at capturing statistical patterns within their native molecular alphabet, cellular biology does not operate as isolated linguistic streams. Cellular phenotype is governed by the Central Dogma of Molecular Biology—an interconnected, multi-tier regulatory cascade in which chromatin architecture controls DNA accessibility, transcription yields structured and spliced mRNA transcripts, and ribosomal translation synthesizes functional three-dimensional proteins (Alberts et al., Molecular 189–98; Nelson et al., secs. 3.1, 27.1). In response to the inherent limitations of single-modality encoders, an emerging frontier of research has sought to achieve multimodal biological fusion by combining representations from DNA, RNA, and protein models (Mollaysa et al. 1–4; Garau-Luis et al. 1–6; He et al. 942–46).

Despite impressive initial performance claims, current multimodal biological fusion literature suffers from fundamental organizational, conceptual, and methodological shortcomings. First, existing studies frequently conflate computational sequence-derived views with true independent biological measurements, translating a single mRNA into artificial DNA and protein strings without providing new physical context (Mollaysa et al. 2). Second, the literature universally lacks rigorous baseline controls: foundation model embeddings are rarely benchmarked against simple, low-cost sequence statistics like length and nucleotide composition, masking whether complex architectures have

learned genuine biological syntax or merely act as high-dimensional character counters. Third, existing frameworks operate under the unverified assumption that cross-modal representations must be forced into geometric alignment, ignoring the reality that biological modalities are valuable precisely because they encode orthogonal, non-overlapping constraints. Finally, current fusion pipelines rely predominantly on quadratic cross-attention mechanisms, inducing severe memory bottlenecks that force arbitrary sequence truncation and destroy regulatory context.

This review tells the story of how biological sequence modeling reached this critical juncture and outlines our framework to resolve these shortcomings. In accordance with established methodological guidelines for synthesizing scientific literature (Leedy and Ormrod 74–77), the review is structured around five successive subproblems: (1) Section 2 traces the transition from classical biophysical heuristics to unimodal foundation models, identifying the structural and regulatory blind spots that prevent any single encoder from capturing the full phenotypic landscape; (2) Section 3 evaluates the initial wave of biological fusion architectures, categorizing how recent studies combine representations across molecular modalities; (3) Section 4 diagnoses the four fatal flaws in the current literature: the sequence-derived multi-view fallacy, the omission of compositional baselines, the representation alignment paradox, and architectural scaling bottlenecks; (4) Section 5 details how our capstone research directly addresses these gaps through composition-residualized representation diagnostics, an alignment-free token-level fusion framework, linear-time Coupled Mamba architectures on a unified codon grid ($T = L/3$), and dual-regime empirical benchmarking; and (5) Section 6 synthesizes the reviewed literature into an actionable contribution matrix and pre-registered hypotheses.

2. The Evolution of Biological Sequence Modeling: From Biophysical Heuristics to Unimodal Encoders

Biophysical Determinants of Sequence Behavior

Before the advent of deep sequence models, predictive biology established that post-transcriptional regulation is governed by physical and statistical features embedded within primary sequences. In messenger RNA degradation, codon optimality—the efficiency and accuracy with which transfer RNAs decode specific trinucleotide codons during ribosomal elongation—was identified as a primary driver of transcript stability (Presnyak et al. 1111–14). Synonymous codons, though encoding the identical amino acid, exhibit starkly divergent translation kinetics: optimal codons accelerate ribosomal transit and stabilize the transcript, whereas non-optimal codons trigger ribosome stalling, recruitment of deadenylation complexes, and accelerated transcript clearance (Nelson et al., sec. 27.1; Diez et al. 1–3). Beyond codon composition, transcript length, overall GC content, wobble-position GC content (GC3), secondary structure folding energy near the translation initiation site, and binding motifs for RNA-binding proteins in untranslated regions (UTRs) collectively dictate stability and translation efficiency (Agarwal and Kelley 245; Nieuwkoop et al. 2363–67).

In parallel, at the genomic level, steady-state transcript expression depends on distant regulatory architecture: enhancers, promoters, and insulator CTCF-binding elements whose access is mediated by chromatin packing and histone modifications (Alberts et al., Molecular 189–98, 334). Splicing

removes introns to generate alternative transcript isoforms, and untranslated regions mediate post-transcriptional localization and persistence (Alberts et al., Molecular 339–40). However, classical biophysical modeling required manual feature engineering that could not scale to the non-linear, combinatorial interactions spanning full-length transcripts and multi-kilobase regulatory regions.

The Unimodal Foundation Model Revolution

To bypass manual feature extraction, the field turned to self-supervised learning, resulting in three distinct families of unimodal biological foundation models: (1) Genomic DNA Encoders, including the Nucleotide Transformer family (Dalla-Torre et al. 287–91), DNABERT-2 (Zhou et al. 1–6), and HyenaDNA (Nguyen et al., 'HyenaDNA' 43177–82), pretrain transformer or long-convolution backbones on vast genomic assemblies. Nucleotide Transformer utilizes 6-mer tokenization to capture local motif grammar; DNABERT-2 implements byte-pair encoding (BPE) to construct variable-length nucleotide vocabularies; and HyenaDNA uses single-nucleotide sub-quadratic convolutions to extend receptive fields up to 1 million base pairs. (2) Transcriptomic RNA and Codon Encoders specialize in RNA sequences. RNA-FM (Chen et al. 1–8) and RiNALMo (Penić et al. 1–9) train single-nucleotide masked language models on hundreds of millions of non-coding RNA sequences from RNAcentral, excelling at capturing structural RNA base-pairing. Conversely, models such as CodonBERT (S. Li et al. 1027–30), CaLM (Outeiral and Deane 170–74), and mRNA-FM (ml4bio) pretrain directly on protein-coding sequences (CDS) using in-frame codon tokenization, capturing organism-specific codon biases, translation elongation efficiency, and transcript stability dynamics. (3) Protein Language Models, represented preeminently by ESM-2 (Lin et al. 1123–27) and ProtBERT (Elnaggar et al. 7112–18), train masked language objectives over tens of millions of evolutionary diverse amino acid sequences in UniRef. Across scaling regimes spanning 8 million to 15 billion parameters, ESM-2 internalizes the physicochemical laws of protein folding, secondary structure propensities, and residue-residue co-evolutionary contacts without observing physical coordinates.

Biological Capabilities and Inherent Blind Spots of Unimodal Models

Despite their individual computational power, Table 1 demonstrates that each unimodal encoder family is fundamentally blind to critical segments of the Central Dogma regulatory cascade.

Modality Family	Exemplar Models	Tokenization Scheme	Critical Biological Strengths	Inherent Modality Blind Spots
Genomic DNA	Nucleotide Transformer, DNABERT-2, HyenaDNA	6-mer, BPE, or Single Nucleotide	Long-range upstream/downstream regulation; chromatin accessibility; promoter/enhancer grammar.	Blind to mature splicing events, UTR regulatory dynamics, and post-transcriptional mRNA kinetics when given CDS alone.
Transcriptomic RNA / Codon	RNA-FM, CaLM, CodonBERT, mRNA-FM	Codon triplets or Single Nucleotide	Synonymous codon optimality; ribosome decoding rates; secondary structure in UTRs.	Blind to long-range chromatin architecture; lacks explicit representations of 3D protein tertiary fold and

				co-evolutionary residue contacts.
Proteomic	ESM-2 (8M–15B), ProtBERT	Single Amino Acid Residues	3D structural propensities; evolutionary contact maps; enzyme catalytic active sites.	Completely blind to synonymous codon variation (translating degenerate codons to one residue strips away mRNA stability and translation rate signals).

This biological segregation exposes an unmistakable conclusion: no single molecular modality contains sufficient information to predict complex downstream phenotypes. A protein encoder receiving the translated amino acid sequence cannot distinguish between an optimized, highly stable mRNA transcript and a rapidly degraded, non-optimal synonymous transcript (Nieuwkoop et al. 2363–66). Conversely, a DNA or RNA encoder struggles to infer the delicate co-evolutionary residue contacts that dictate the folding stability of the translated enzyme (Lin et al. 1123–25). This fundamental informational divide provided the impetus for multimodal biological fusion.

3. The Emergence of Multimodal Biological Fusion: Architectures and Paradigms

Recognizing that biological modalities possess complementary views of the Central Dogma, researchers recently pioneered multimodal frameworks designed to synthesize nucleic acid and proteomic representations. These efforts can be synthesized into three primary architectural paradigms:

Paradigm 1: Late-Stage, Sequence-Derived Multi-View Fusion

The first paradigm, exemplified by BioLangFusion (Mollaysa et al. 1–5), addresses sequence property prediction by artificially synthesizing three molecular views from a single input sequence. Taking a coding mRNA transcript, BioLangFusion deterministically generates the corresponding DNA sequence (by replacing uracil with thymine) and the translated protein sequence via standard ribosomal translation. It processes these streams through three frozen, independently pretrained unimodal encoders: Nucleotide Transformer v2 (100M) for DNA, RNA-FM for RNA, and ESM-2 (8M) for protein. Because each model operates on a distinct tokenization scale—6-mers, single nucleotides, and amino acid residues—BioLangFusion resamples the representations onto a common spatial grid referenced to protein amino acid coordinates. Specifically, single-nucleotide RNA embeddings are averaged in non-overlapping groups of three (1x3 pooling), while DNA representations are projected through a learned transposed 1D convolution. The authors then evaluate position-wise concatenation, attention-based modality pooling, and dense cross-attention, reporting that fused representations outperformed unimodal baselines across mRNA stability and protein expression benchmarks.

Paradigm 2: Multi-Input Genomic and Transcriptomic Cross-Attention

A more biologically grounded fusion paradigm is embodied by IsoFormer (Garau-Luis et al. 1–9). Rather than deriving artificial sequence views from a single transcript, IsoFormer takes three genuinely distinct biological inputs to predict tissue-specific transcript abundance across 30 human tissues in the Genotype-Tissue Expression (GTEx v8) atlas: (1) Long-range genomic DNA (12 kb to 196 kb centered on the transcription start site), processed by Enformer or Nucleotide Transformer v2; (2) Full-length mature mRNA transcripts (including 5' and 3' UTRs), processed by Nucleotide Transformer v2; and (3) The translated protein amino acid sequence (when applicable), processed by ESM-2 (150M). Long transcripts are cropped from the 5' end to preserve critical 3' UTR regulatory signals, and non-coding transcripts are handled by setting missing protein cross-attention projections to zero. Through stacked multi-head cross-attention and residual feed-forward layers, IsoFormer allows the genomic DNA stream to inform transcript processing and vice versa, reporting a rise in predictive R^2 from 0.36 (RNA alone) to 0.53 (fused DNA + RNA + Protein).

Paradigm 3: Unified Alphabet Pretraining and Directional Transformers

Beyond post-hoc fusion, two alternative paradigms have emerged: Unified Pretraining, exemplified by LucaOne (He et al. 942–48), an 8-layer, 1.8-billion-parameter transformer pretrained simultaneously on nucleic acid and protein sequences using a shared multi-alphabet vocabulary; and Directional Mechanistic Modeling, exemplified by Central Dogma Transformer (CDT v1; Ota 1–15), which enforces an explicit computational pipeline where enhancer-centered DNA embeddings (via frozen Enformer) are queried by RNA embeddings (via scGPT gene tokens), which are subsequently queried by protein representations (via ProteomeLM) to predict CRISPRi perturbation responses.

4. Critical Analysis: Where Previous Research Falls Short

While the aforementioned studies demonstrate that combining models can yield higher benchmark scores, a rigorous scientific evaluation reveals four critical, unaddressed shortcomings that undermine the validity and generalizability of their conclusions.

Shortcoming 1: The Sequence-Derived Multi-View Fallacy

A core conceptual flaw in frameworks like BioLangFusion (Mollaysa et al. 1–4) is the conflation of a computational multi-view transformation with a multimodal biological measurement. Generating DNA and protein strings from a single coding mRNA transcript creates zero new independent information: the translated protein sequence is a deterministic, lossy projection of the RNA coding sequence that discards all synonymous codon identity (Nelson et al., sec. 27.1). If an RNA foundation model has already observed the transcript, feeding the identical translated sequence into ESM-2 does not provide a new physical observation of the cell. Any observed performance gain can stem only from the inductive biases and evolutionary contact priors learned by ESM-2 during UniRef pretraining—not from multimodal biological synergy. Prior studies fail to isolate whether observed improvements are genuinely multimodal or merely an ensemble effect across different pretraining objectives.

Shortcoming 2: The Composition Confounder and Lack of Baseline Controls

The most severe methodological failure in the current literature is the near-total absence of low-cost statistical baseline controls (Hewitt and Liang 2733–37). Biological sequences possess prominent compositional features—transcript length, overall GC content, codon frequencies, and amino acid usage—that correlate strongly with stability and expression. Existing papers attribute high prediction scores entirely to the deep semantic abstractions of foundation models without verifying how much variance is explained by trivial counting statistics. When subjected to linear residualization against an 87-feature baseline (comprising length, GC, GC3, 64 codon frequencies, and 20 amino acid frequencies), empirical evaluations reveal an alarming reality: between 61% and 92.4% of the embedding variance in genomic DNA foundation models (e.g., HyenaDNA at 92.4%, Nucleotide Transformer v2 at 80.4%) is purely linear sequence composition. These multi-million-parameter transformers are largely functioning as overparameterized k-mer counters. Even more strikingly, on the GTEx tissue expression benchmark, four simple transcript-length features achieve an out-of-fold cross-validation R^2 of 0.242—outperforming all twelve frozen biological foundation models combined ($R^2 = 0.239$). By omitting simple linear and compositional controls, previous studies claim deep biological representation learning where simple sequence statistics suffice.

Shortcoming 3: The Representation Alignment Paradox

A widespread assumption in multimodal machine learning is that effective fusion requires cross-modal representations to be geometrically aligned—either by maximizing Centered Kernel Alignment (CKA) or optimizing explicit contrastive losses (e.g., CLIP-style InfoNCE) to pull paired DNA, RNA, and protein embeddings together (Radford et al. 8748–52; Kornblith et al. 3519–24). This assumption represents a fatal misunderstanding of biological representation learning (Tjandrasuwita et al. 59723–28). When raw linear CKA is calculated between DNA, RNA, and protein foundation models, observed similarity appears deceptively high ($0.60 \leq \text{CKA} \leq 0.80$). However, when the 87 compositional features are residualized out of the representations, cross-modal CKA collapses by 52% to 90%. The apparent alignment observed in prior literature is an artifact of shared character composition (both encoders tracking GC content and codon bias). Furthermore, biological modalities are valuable precisely because they capture orthogonal, non-overlapping constraints: protein models capture tertiary fold constraints, while RNA codon models capture translation kinetics. Enforcing an explicit sequence-level alignment loss penalizes representations whenever they diverge, actively forcing the encoders to discard their unique structural signals and collapse onto their lowest common denominator: trivial nucleotide composition. Prior research has failed to recognize that biological fusion requires complementary integration without geometric alignment.

Shortcoming 4: Architectural Scaling Bottlenecks and Evaluation Leakage

Finally, previous fusion implementations suffer from crippling architectural and benchmarking limitations: (1) Quadratic Memory Bottlenecks: Standard multi-head cross-attention scales quadratically, $O(L^2)$, with sequence length (Vaswani et al. 5998–6002). For biological transcripts that frequently span 3,000 to 10,000 nucleotides, cross-attention forces practitioners to severely crop UTRs (as in IsoFormer) or compress tokens into coarse windows, discarding the fine-grained

nucleotide positions where regulatory motifs reside. (2) Uncontrolled Optimization and Arbitrary Layer Selection: BioLangFusion and IsoFormer evaluate models across single random seeds without reporting fold-assignment stability or multi-seed standard deviations. Moreover, they default to extracting representations from the final encoder layer, directly ignoring established empirical findings that intermediate layers of genomic foundation models consistently outperform final layers on downstream molecular tasks (Dalla-Torre et al. 288, 294; Alain and Bengio 1–6). (3) Homology and Partition Leakage: Published benchmarks frequently split datasets randomly rather than separating training and testing sets by gene family or sequence homology, creating substantial risk of data leakage where homologous sequence patterns artificially inflate test performance (Kapoor and Narayanan 100804).

5. Bridging the Gaps: Where We Plan to Take the Research to Add to the Body of Knowledge

To resolve these fundamental shortcomings, our capstone research introduces a disciplined, four-stage experimental framework that bridges the gap between foundation model representations and biophysical sequence regulation.

Contribution 1: Establishing Composition-Controlled Representation Diagnostics (Stage 1)

To ensure that learned representations reflect genuine biological syntax rather than trivial character counts, we establish a standardized evaluation protocol. We define an 87-feature linear baseline matrix F capturing overall GC, wobble-position GC3, log-transformed length, 64 triplet codon frequencies, and 20 amino acid frequencies. For every encoder X in $\mathbb{R}^{(N \times D)}$, we compute its in-sample composition share via multivariable ordinary least squares residualization: $X_{\text{residual}} = X - X_{\text{hat}}$, and $R^2_{\text{comp}} = 1 - (\|X_{\text{residual}}\|_F^2 / \|X - X_{\text{mean}}\|_F^2)$. By evaluating downstream decodability with Standardized Ridge probes using Leave-One-Out Cross-Validation (LOOCV) and 5-fold cross-validation across both raw embeddings (X) and residualized embeddings (X_{residual}), our framework isolates the true non-compositional biological signal contributed by each foundation model.

Contribution 2: Reframing Alignment and Rejecting Destructive Contrastive Losses (Stage 2)

We formally test the representation alignment hypothesis across twelve frozen foundation models spanning DNA (Nucleotide Transformer v2, DNABERT-2, HyenaDNA), RNA (RNA-FM, RiNALMo, CaLM, mRNA-FM), and protein (ESM-2 across 8M, 35M, and 150M; ProtBERT). Using linear CKA, mutual k -nearest-neighbor overlap ($k=10$), and canonical correlation analysis (CCA) before and after residualization, we demonstrate that true biological representations are inherently orthogonal. Crucially, this diagnosis establishes a pivotal design decision: we explicitly drop the sequence-level contrastive alignment loss (e.g., CLIP-style objectives) from our fusion pipeline. Instead of forcing distinct biological modalities to mimic each other, our framework preserves their native geometries, directing the model to learn multimodal interactions exclusively at the downstream fusion layer.

Contribution 3: Pioneering Linear-Time Token-Level Fusion via Coupled Mamba (Stage 3)

To overcome the quadratic memory bottleneck of cross-attention without discarding sequence context, we propose a linear-time, $O(L)$, token-level fusion architecture: (1) Unified Codon Spatial Grid ($T = L/3$): Because coding sequences naturally adhere to triplet reading frames, we align all sequence streams to the codon coordinate space. Single-nucleotide RNA models are pooled via 1D convolutions with kernel size 3 and stride 3; multi-nucleotide DNA representations are resampled to match the $T = L/3$ temporal dimension; and protein amino acid tokens align natively (1:1) to the codon grid. (2) Coupled Mamba State-Space Duality: Rather than computing pairwise all-to-all attention matrices ($O(L^2)$), we adapt the Coupled Mamba state-space architecture (W. Li et al. 59808–15) to biological sequence modeling. Each molecular stream maintains an independent hidden state $h_m(t)$, and inter-modal cross-talk is achieved by coupling the state transition matrices: $h_m(t) = A_m h_m(t-1) + B_m x_m(t) + \sum_{k \neq m} C_{\{k \rightarrow m\}} h_k(t-1)$. This formulation enables bidirectional, token-level cross-modal communication across full-length transcripts in strictly linear time, $O(L)$, and constant memory, allowing our model to incorporate extensive UTR contexts that prior cross-attention models are forced to crop. We benchmark this priority architecture against IsoFormer-style cross-attention and simple concatenation baselines.

Contribution 4: Rigorous Multi-Task Benchmarking across Divergent Biological Regimes (Stage 4)

Finally, we evaluate our multimodal framework across two fundamentally divergent biological prediction tasks to establish empirical boundary conditions: (1) Regime A: CDS-Driven mRNA Stability (CodonBERT Benchmark), evaluated on 981 curated mRNA sequences, where variance is dominated by intracellular codon optimality, translation elongation rates, and wobble-position GC3 content; and (2) Regime B: Regulatory Context-Driven Transcript Expression (GTEx Isoformer Benchmark), evaluated on tissue-specific steady-state transcript expression across 30 human organs, where variance is governed by transcript length, transcriptional initiation, and promoter/enhancer accessibility. All evaluations are conducted under pre-registered hypotheses (H1–H5), gene-separated and homology-partitioned splits to prevent data leakage, repeated-seed partitioning ($N=10$ assignments) to report true fold variance, and comprehensive ablation suites (systematically removing individual modalities, stripping composition controls, and probing intermediate versus final layers).

6. Summary: Gaps in the Literature and Proposed Contribution

In accordance with the synthesizing principles of academic literature reviews (Leedy and Ormrod 74–77), Table 2 synthesizes the state of the literature, contrasting the specific limitations of prior research with the concrete contributions of our capstone project.

Dimension	State of Prior Literature (BioLangFusion, IsoFormer, LucaOne)	Critical Shortcoming in Existing Work	Our Proposed Capstone Contribution
Input Framing	Synthetic multi-view extraction from single mRNA	Pseudo-multimodality; fails to distinguish learned inductive	Explicitly separates sequence-derived views (CDS mRNA) from

	transcripts (BioLangFusion).	priors from new biological observations.	true multi-input genomic context (GTEx).
Baseline Controls	Reports raw correlation (r , ρ , R^2) without simple statistical baselines.	Ignores that up to 92.4% of model variance is simple k-mer counting; 4 length features beat 12 FMs on GTEx.	Establishes standardized 87-feature composition controls and reports residualized performance (X_{residual}).
Alignment Paradigm	Assumes cross-modal representations must be geometrically aligned via contrastive loss.	Forcing sequence alignment destroys specialized orthogonal signals, collapsing models onto shared GC/codon noise.	Decouples geometry from fusion; eliminates destructive alignment losses in favor of downstream complementarity.
Fusion Mechanism	Quadratic multi-head cross-attention $O(L^2)$ or coarse global pooling.	Quadratic memory bloat forces aggressive sequence truncation, discarding critical regulatory UTRs.	Introduces linear-time $O(L)$ Coupled Mamba on a unified codon grid ($T = L/3$), preserving full sequence context.
Layer Selection	Exclusively extracts representations from the final transformer layer.	Contradicts genomic empirical evidence showing final layers underperform intermediate states.	Evaluates layer-wise CKA and systematic intermediate-layer decodability across all 12 frozen encoders.
Evaluation Rigor	Single-seed splits; unverified homology leakage; task-specific cherry-picking.	Conflates partition variance with true model superiority; risks generalization failure.	Multi-assignment fold testing ($N=10$), strict gene/homology holdout partitions, and full ablation hierarchy.

Formal Pre-Registered Hypotheses

This literature synthesis directly motivates the five empirical hypotheses governing our study: (H1) The Compositionality Ceiling: A significant fraction ($\geq 60\%$) of unimodal embedding variance across genomic DNA and RNA foundation models is attributable to linear sequence composition ($R^2_{\text{comp}} \geq 0.60$), and simple sequence baselines (length, GC, codon frequencies) will match or exceed unimodal FM zero-shot probe performance on unadjusted targets. (H2) The True-Signal Complementarity Hypothesis: On composition-residualized stability targets (y_{residual}), combining a codon-aware RNA encoder (CaLM) with an evolutionary protein encoder (ESM-2) will yield statistically significant predictive gains ($\Delta R^2 \geq 0.05$, $p < 0.01$) over any single constituent encoder. (H3) The Alignment-Destruction Principle: Optimizing an explicit sequence-level contrastive alignment loss (e.g., InfoNCE) across frozen encoder projections will degrade downstream multimodal prediction accuracy compared to unconstrained fusion. (H4) State-Space Computational Efficiency: Token-level Coupled Mamba will achieve predictive parity with, or superiority over, multi-head cross-attention while maintaining strictly linear $O(L)$ memory scaling across full-length transcripts. (H5) Divergent Regime Selectivity: Multimodal gain profiles will diverge sharply across biological regimes: codon-aware RNA-protein fusion will dominate CDS-driven mRNA stability, whereas long-range genomic DNA context and length features will dominate GTEx tissue expression.

By executing this experimental framework, our capstone project moves biological machine learning past overparameterized character counting, delivering a rigorous, computationally scalable methodology for choosing and fusing biological foundation models.

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